Prompt Engineering – Assignment 1 Answers

Total Marks: 45 | Part A: 15 Marks | Part B: 30 Marks

Part A – Prompt Design Exercises (15 Marks)

Q1. Vague Prompt vs Structured Prompt (5 Marks)

Vague Prompt:

Summarize this news article.

Sample news article (for testing):

A technology company has launched an AI-powered healthcare assistant that helps doctors analyze patient records and identify potential health risks. The system aims to reduce administrative workloads and improve decision-making. However, experts have raised concerns about patient privacy, data security, and the reliability of AI-generated recommendations.

Improved Prompt Using Six Components

Component	Description
Role	You are an experienced technology news analyst.
Task	Summarize the provided news article.
Input	The news article provided above.
Instruction	Identify the main development, benefits, and concerns. Avoid adding unsupported information.
Context	The summary is intended for nontechnical readers.
Output Format	Provide three bullet points, each under 25 words.

Complete Improved Prompt:

PROMPT

You are an experienced technology news analyst. Summarize the following article for readers without a technical background. Identify the main development, benefits, and concerns. Do not introduce information absent from the article. Present the summary in three bullet points, each under 25 words.

Article: A technology company has launched an AI-powered healthcare assistant that helps doctors analyze patient records and identify potential health risks. The system aims to reduce administrative workloads and improve decision-making. However, experts have raised

concerns about patient privacy, data security, and the reliability of AI-generated recommendations.

Explanation: The vague prompt provides limited guidance, whereas the improved prompt specifies the role, task, input, instructions, context, and output format. This produces a clearer, more focused, and structured summary.

Q2. Few-Shot Prompt for Customer Review Classification (4 Marks)

Prompt:

Classify customer reviews into Positive, Negative, or Neutral based on the sentiment expressed.

Examples:

Review: "The product quality is excellent, and I love it."

Sentiment: Positive

Review: "The product stopped working after two days. Very disappointing."

Sentiment: Negative

Review: "The product arrived yesterday and matches the description."

Sentiment: Neutral

Now classify the following reviews. Return only the sentiment label.

Review 1: "The delivery was fast, and the product works perfectly."

Review 2: "The packaging was damaged, and customer support was unhelpful."

Expected Responses:

Test Review	Classification
Review 1	Positive
Review 2	Negative

Explanation: Few-shot prompting provides labeled examples before asking the model to classify new inputs. The examples demonstrate the expected classification behavior and output format.

Q3. Chain-of-Thought vs Zero-Shot Prompting (3 Marks)

Word Problem:

A shop sells notebooks for ₹40 each. A customer buys 5 notebooks and receives a 10% discount on the total bill. How much does the customer pay?

Zero-Shot Prompt:

A shop sells notebooks for ₹40 each. A customer buys 5 notebooks and receives a 10% discount. How much does the customer pay?

Expected Response:

₹180.

Chain-of-Thought Prompt:

A shop sells notebooks for ₹40 each. A customer buys 5 notebooks and receives a 10% discount on the total bill. Calculate the answer step by step, showing the total cost, discount amount, and final payable amount.

Expected Response:

1. Cost of one notebook = ₹40.
2. Cost of 5 notebooks = ₹40 × 5 = ₹200.
3. Discount = 10% of ₹200 = ₹20.
4. Final amount = ₹200 − ₹20 = ₹180.

Comparison:

Zero-Shot	Chain-of-Thought
Gives the final answer directly	Shows intermediate calculation steps
Short and concise	More explanatory
₹180	₹180

Conclusion: Both approaches produce the same answer, but the step-by-step prompt makes the calculation easier to verify.

Q4. Self-Consistency Prompting (3 Marks)

Reasoning Prompt:

A train travels 60 km in the first hour, 80 km in the second hour, and 100 km in the third hour. What is its average speed for the entire journey? Show the calculation and provide the final answer in km/h.

Sampling Configuration:

- Temperature: 0.7
- Number of runs: 5
- Prompt: Identical for all runs

Illustrative Results:

Run	Answer	Calculation
1	80 km/h	240 ÷ 3
2	80 km/h	(60 + 80 + 100) ÷ 3
3	80 km/h	240 ÷ 3
4	80 km/h	Total distance / total time
5	80 km/h	(60 + 80 + 100) / 3

Majority Vote:

80 km/h

5 of 5 illustrative answers

Final Answer: 80 km/h.

Explanation: Self-consistency generates multiple responses using a nonzero temperature and selects the most frequent final answer. It can improve reliability on reasoning tasks.

Part B – Multiple Choice Questions (30 Marks)

Here are the correct answers for all 30 questions.

Q.No.	Correct Option	Answer
1	B	Role
2	C	Instruction
3	D	Task states what must be accomplished; Instruction specifies how to perform it
4	A	Output Format
5	C	Input
6	B	Context
7	D	Output Format
8	A	Instruction describes how to perform the task; Output Format describes how the answer should look
9	B	Instructions and context sent to the model
10	C	Total tokens the model handles at once: prompt plus output
11	A	Maximum length of generated output
12	D	Temperature, top-p, max_output_tokens
13	B	Strongly peaked, more predictable output
14	C	Flatter, more varied output
15	A	Probability-weighted selection from the top 3 tokens
16	B	Top tokens selected by cumulative probability P

17	C	Top-N uses token count; Top-P uses cumulative probability
18	B	Designing inputs for LLMs to obtain desired responses
19	C	Uses pretrained knowledge without examples
20	D	Provides a few examples to guide output
21	B	Few-shot prompting
22	C	Chain-of-Thought prompting
23	B	11 apples
24	D	Reason → Action → Observation → Response
25	A	Needs external search or tools to obtain live data
26	C	Multiple runs with temperature above 0, selecting the most common answer
27	A	Your own code or process, such as majority voting
28	A	Computationally expensive due to multiple runs
29	C	Summarize in three bullet points, each under 20 words
30	D	Role-based prompting